



ALLEN
CAREER INSTITUTE
KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)

REVIEW TEST

28-02-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

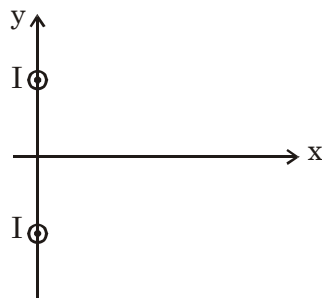
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Two infinitely long parallel wires are a distance d apart and carry equal parallel currents I in the same direction as shown in the figure. If the wires are located on y -axis (normal to x - y plane) at $y = \frac{d}{2}$ and $y = -\frac{d}{2}$, then the magnitude of x -coordinate of the point on x -axis where the magnitude of magnetic field is maximum is (Consider points on x -axis only)



(A) $\sqrt{2}d$

(B) $\frac{d}{\sqrt{2}}$

(C) $2d$

(D) $\frac{d}{2}$

Space for Rough Work

2. A projectile is fired vertically upward with an initial velocity u at $t=0$. After an interval of T seconds a second projectile is fired vertically upwards, also with initial velocity u .

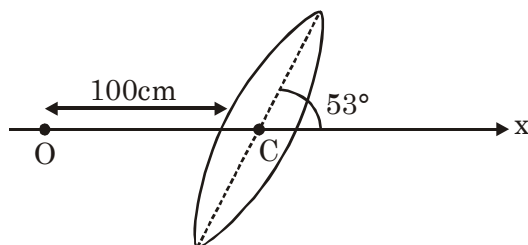
(A) They meet at time $t = \frac{u}{g} - \frac{T}{2}$ and at a height $\frac{u^2}{2g} - \frac{gT^2}{8}$.

(B) They meet at time $t = \frac{u}{g} + \frac{T}{2}$ and at a height $\frac{u^2}{2g} + \frac{gT^2}{8}$.

(C) They meet at time $t = \frac{u}{g} + \frac{T}{2}$ and at a height $\frac{u^2}{2g} - \frac{gT^2}{8}$.

(D) None of these.

3. As shown in figure, a thin convex lens of focal length 20 cm is inclined at an angle of 53° with x -axis. A point object is placed on the x -axis at a distance of 100 cm from the centre of the lens. Taking the origin as the centre of lens, find the x -coordinate of the image.



(A) 26.7 cm

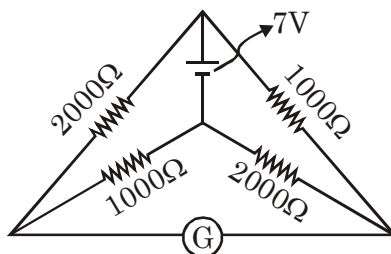
(B) 25 cm

(C) 33.3 cm

(D) 20 cm

Space for Rough Work

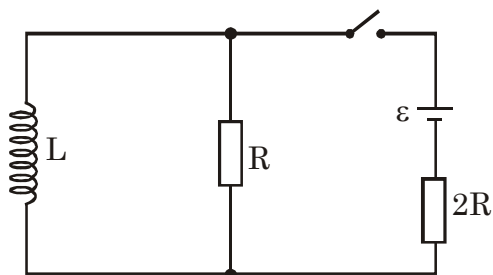
4. In the following circuit the values of resistance in various arms are as shown. The galvanometer has a current sensitivity of $\frac{1\text{mm}}{100\mu\text{A}}$ and an internal resistance of $1000\ \Omega$. Calculate the deflection in galvanometer.



- (A) Zero (B) 10 mm (C) 20 mm (D) 30 mm
5. We released four wooden floating rafts on a river from port A. There is a boat whose speed relative to river is 4 m/s irrespective of whether it is connected to a raft or not. At a time, only one raft can be connected to the boat. The rafts have to be carried to a port B at a distance of 27 km downstream, the flow rate of the river is 2 m/s. What is the approx. minimum time all rafts can be delivered to port B? Neglect the time to tie the raft to the boat and releasing it. Also neglect the dimensions of boat, raft and the connecting rope in comparison with 27km.
- (A) 3 hrs. 45 min. (B) 2 hrs. 25 min. (C) 4 hrs. 35 min. (D) 1 hrs. 12 min.

Space for Rough Work

6. In the diagram shown in Figure, all the elements can be considered ideal, the parameters of the elements shown in the figure. Before key is connected, there was no current. Key is closed for some time and then opened. It turned out that the magnitude of current through the resistor resistance R immediately before opening the key and immediately after opening the key is the same. How much heat is dissipated in the circuit after opening the key?



- (A) $\frac{L\epsilon^2}{25R^2}$ (B) $\frac{L\epsilon^2}{20R^2}$ (C) $\frac{L\epsilon^2}{50R^2}$ (D) $\frac{L\epsilon^2}{30R^2}$

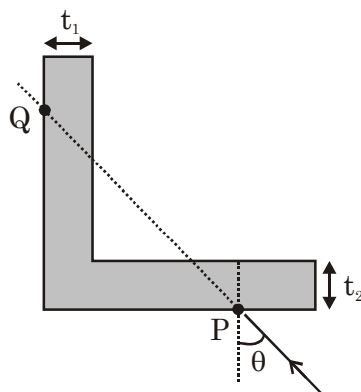
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
7. A stick of total length L is taken. A certain amount of charge is uniformly distributed along its length. Let the resultant charge on the stick be Q and its mass be M . The entire structure is spun about the centre about an axis perpendicular to the stick with an angular velocity ω . There exists a uniform magnetic field B parallel to the axis of rotation. If the rotational kinetic energy is K , and energy of interaction of the structure with magnetic field is E ,
- (A) $E = -\frac{QBK}{M\omega}$
- (B) $E = -\frac{QBK}{2M\omega}$
- (C) To rotate the structure so that the angular velocity vector becomes perpendicular to the magnetic field, the external agent has to do positive work.
- (D) To rotate the structure so that the angular velocity vector becomes perpendicular to the magnetic field, the external agent has to do negative work.

Space for Rough Work

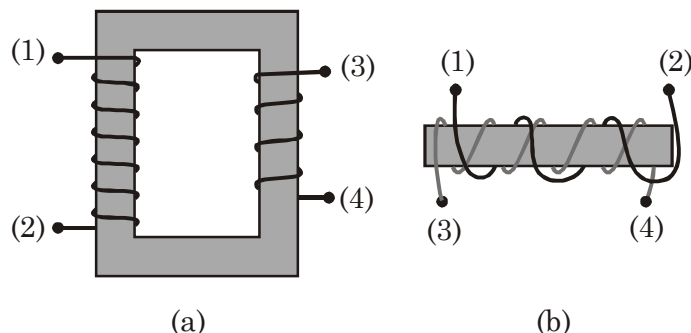
8. Two glass slabs of thickness t_1 and t_2 are joined to make L shape as shown in the figure. The refractive indices of both glass slabs are same. A light ray is incident at point 'P' with an angle of incidence ' θ ' as shown in the figure. It is assumed that refraction takes place from both the glass slabs. Then choose the **CORRECT** option(s) :



- (A) emergent ray passes from Q if $t_1 = 2t_2$, $\theta = 30^\circ$.
 (B) emergent ray passes from Q if $t_1 = 2t_2$, $\theta = 45^\circ$.
 (C) emergent ray passes above the 'Q' if $t_2 > t_1$, $\theta = 45^\circ$.
 (D) emergent ray passes below the 'Q' if $t_2 < t_1$, $\theta = 45^\circ$.

Space for Rough Work

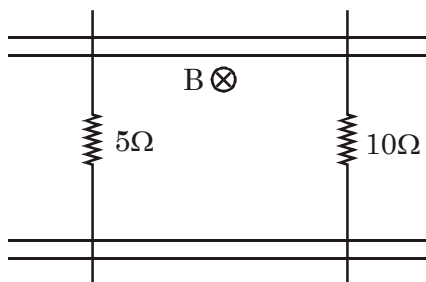
9. For each of the transformer in figure, assume that current in primary coil is going from terminal-1 to terminal-2 and increasing with time. Choose the correct statement(s) about the polarity of induced emf in secondary coil.



- (A) For (a), $V_3 > V_4$ (B) For (a), $V_3 < V_4$ (C) For (b), $V_3 > V_4$ (D) For (b), $V_3 < V_4$
10. A proton of charge 'e' and mass 'm' enters a uniform and constant magnetic field $\vec{B} = B\hat{i}$ with an initial velocity $\vec{V} = V_{OX}\hat{i} + V_{OY}\hat{j}$. Which of the following will be correct at any later time 't':
- (A) x-component of velocity of the proton will be $V_{OX} \cos\left(\frac{qBt}{m}\right)$
- (B) y-component of velocity will be $V_{OY} \cos\left(\frac{qBt}{m}\right)$
- (C) z-component of velocity will be $V_{OY} \sin\left(\frac{qBt}{m}\right)$ along $(+\hat{k})$ direction.
- (D) x-component of velocity will remain unchanged.

Space for Rough Work

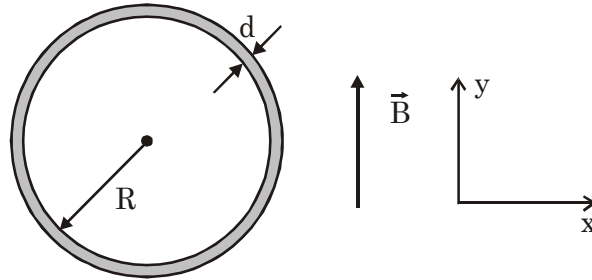
11. A pair of parallel conducting rails lie at right angle to a uniform magnetic field of 2.0 T as shown in the figure. Two resistors 10Ω and 5Ω are to slide without friction along the rail. The distance between the conducting rails is 0.1 m. Then :



- (A) Induced current = $\frac{1}{150}$ A directed clockwise if 10Ω resistor is pulled to the right with speed 0.5 ms^{-1} and 5Ω resistor is held fixed.
- (B) Induced current = $\frac{1}{300}$ A directed anti-clockwise if 10Ω resistor is pulled to the right with speed 0.5 ms^{-1} and 5Ω resistor is held fixed.
- (C) Induced current = $\frac{1}{300}$ A directed clockwise if 5Ω resistor is pulled to the left at 0.5 ms^{-1} and 10Ω resistor is held at rest.
- (D) Induced current = $\frac{1}{150}$ A directed anti-clockwise if 5Ω resistor is pulled to the left at 0.5 ms^{-1} and 10Ω resistor is held at rest.

Space for Rough Work

12. A thin walled nonmagnetic spherical shell with wall thickness d and radius R ($R \gg d$) is placed in a uniform magnetic field $\vec{B} = B_0 \hat{j}$, which slowly changes as $\frac{dB_0}{dt} = \alpha$, where α is a positive constant. Electrical resistivity of the material of the shell is ρ . Then choose the **CORRECT** option(s) :



- (A) Induced magnetic moment of the shell is towards $+\hat{j}$ direction.
- (B) Induced magnetic moment of the shell is towards $-\hat{j}$ direction.
- (C) Induced magnetic moment of the shell is $\frac{3\pi}{2\rho} dR^4 \alpha$.
- (D) Induced magnetic moment of the shell is $\frac{2\pi}{3\rho} dR^4 \alpha$.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

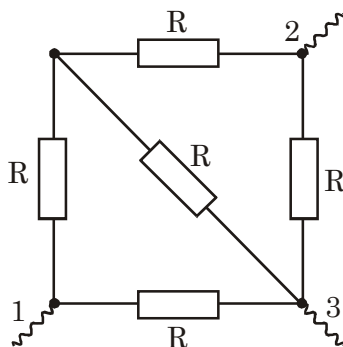
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

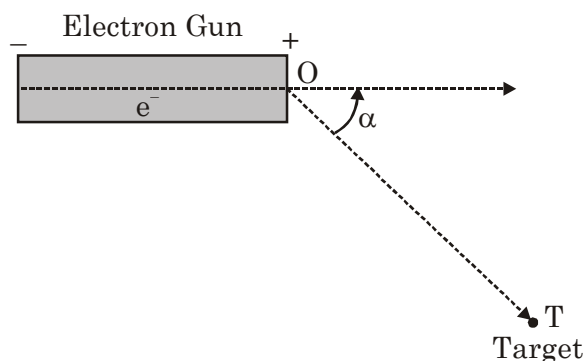
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- The resistances in a burner of an electric table stove are connected together in an arrangement shown in figure. This arrangement is connected to the supply mains at points 1 and 2, and, after a certain time, 500 grams of water are heated to the boiling point. How much water (in gm) can be heated to the boiling point in the same time interval when the arrangement of resistances is connected to the supply mains at points 1 and 3? The initial temperature of the water is the same in both cases. Neglect all heat losses.

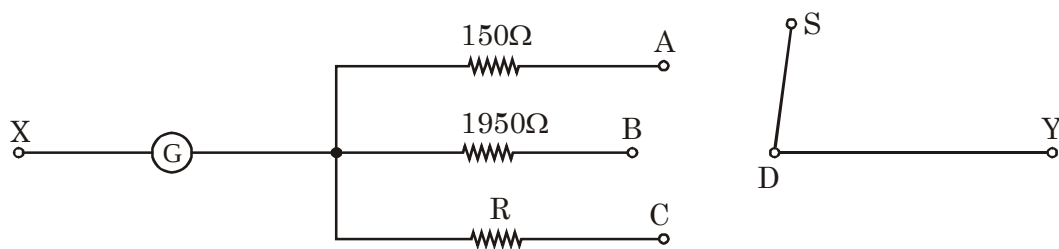


Space for Rough Work

2. Electrons are accelerated by an electron gun. As the electrons exit the gun they are deflected by a uniform magnetic field of fixed magnitude directed into the page. The electrons hit a small (point-like) target. An experimentalist noticed that if the magnetic field is directed along the line OT then also the electrons hit the target. For what minimum angle α is this possible? Fill the value of $\tan \alpha$. Assume that all electrons have the same initial velocity.

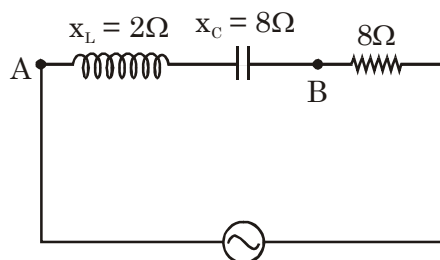


3. In the circuit shown, When the switch S connects points A and D, the galvanometer deflects full-scale when the potential difference between X and Y is 100 V. When the switch S connects points B and D, the galvanometer deflects full-scale when the potential difference between X and Y is 1000 V. What value of R (in Ω) is needed to cause the galvanometer to deflect full-scale when the potential difference between X and Y is 10 kV when the switch S connects C to D ? Fill value of $\frac{R}{10}$.



Space for Rough Work

4. A wire of 5Ω resistance and 20 m length is wound into a coil of radius 4cm. A magnetic field of 2.4 T is perpendicular to the plane of the coil. If the current that forms the magnetic field is switched to the opposite direction, what is the total charge (in C) that passes through the wire coil ?
5. An inductor ($x_L = 2\Omega$) a capacitor ($x_C = 8\Omega$) and a resistance (8Ω) is connected in series with an A.C. source. The voltage output of A.C. source is given by $v = 10 \cos 100\pi t$ (volts). If the instantaneous potential difference between A and B is x when it is half of the voltage output from source at that instant then what is the value of x (in Volts) :



6. A series R-C combination is connected to an AC voltage of frequency $f = \frac{250}{\pi}$ Hz. If the impedance of the R-C circuit is $R\sqrt{5}$, the time constant (in millisecond) of the circuit is.

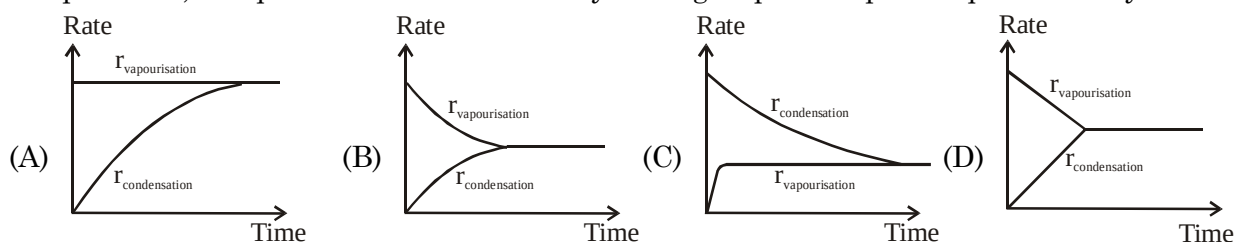
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
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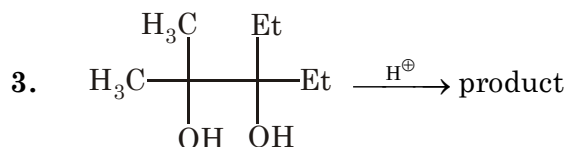
1. Which of following curve represent equilibrium of vapour with its pure liquid at constant temperature, if equilibrium is achieved by taking vapour of pure liquid initially -



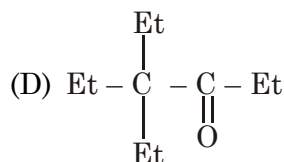
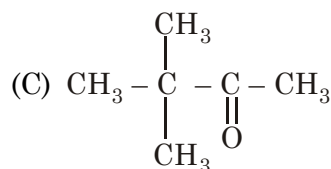
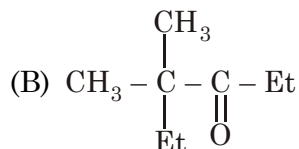
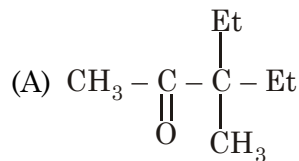
2. Which of the following statement is **CORRECT**?

- (A) :CH_2 is relatively more stable than :CCl_2 and :CF_2
 (B) CHF_3 is more acidic than CHCl_3
 (C) $\text{R}_3\text{C-O-H}$ is more acidic than $\text{R}_3\text{Si-O-H}$
 (D) $\text{B}_3\text{O}_6^{-3}$ and $\text{B}_3\text{N}_3\text{H}_6$ are aromatic in nature

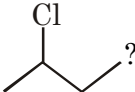
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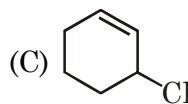
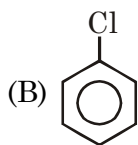
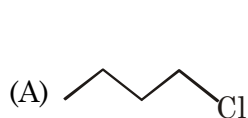


Major product of the reaction is :



4. 2.08 kg of an equimolar mixture of ferrous oxalate (MW = 144) and ferric oxalate (MW = 376) require X moles of $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium for complete oxidation. Then 'X' is-
- (A) 16/3 (B) 6 (C) 32/3 (D) 10/3
5. Which of the following order is **NOT** correct with respect to the given property?
- (A) $\text{NaF} < \text{NaCl} < \text{NaBr} < \text{NaI}$: Ionic character
- (B) $\text{SrC}_2\text{O}_4 < \text{BaC}_2\text{O}_4 < \text{BeC}_2\text{O}_4$: Solubility in water
- (C) $\text{Ca}_3\text{N}_2 < \text{Mg}_3\text{N}_2 < \text{Be}_3\text{N}_2$: Thermal stability
- (D) $\text{AgI} < \text{AgBr}$: covalent character

6. Which of the following alkyl halide give faster $\text{S}_{\text{N}}1$ reaction than ?

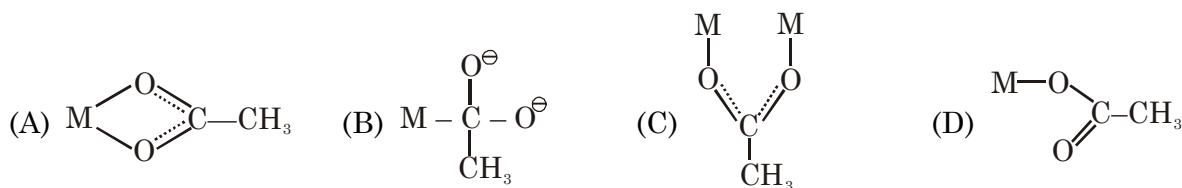


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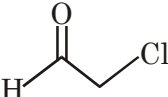
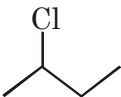
SECTION-I(ii) : (Maximum Marks: 24)

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Pure benzene has a vapour pressure 900 torr and pure toluene has a vapour pressure 360 torr. If a solution is prepared by mixing 39gm benzene and 46 gm toluene then
- (A) Vapour pressure of its distillate will be 745.714 torr
 (B) Vapour pressure of initial solution will be 630 torr
 (C) Mole % of benzene in vapour phase of solution will be $\frac{500}{7}\%$
 (D) Mole % of toluene in its distillate will be 50%.
8. In sodium acetate, acetate ion show distinct ionic behaviour but in complex salt it behave as a ligand. Which of the following is possible ways of bonding of acetate with metal.



Space for Rough Work

9. trans-2-butene $\xrightarrow{\text{Br}_2/\text{CCl}_4}$ Product
Identify correct statement(s) regarding major product?
(A) Product is optically inactive
(B) Racemic mixture is formed as major product
(C) Meso product is formed as major product
(D) Three isomers are possible in major product
10. An atom of U^{238} disintegrates by a series of α -decays and β -decays until it becomes Pb^{206} which is stable. How many of the following ten nuclides may be formed from the series of disintegration starting at U^{238} .
 U^{235} , U^{234} , Ac^{228} , Ra^{224} , Rn^{224} , Rn^{220} , Po^{215} , Po^{212} , Pb^{212} , Pb^{211}
(A) 1 (B) 6 (C) 7 (D) 4
11. If the molecular axis is Z-axis, for which of the following molecular orbitals, energy is changed in N_2 molecule due to s-p mixing?
(A) $\pi 2p_x$ (B) $\sigma^* 2s$ (C) $\pi^* 2p_y$ (D) $\sigma 2p_z$
12. Which of the following alkyl halide give faster $\text{S}_{\text{N}}2$ reaction than Et-Cl
(A) $\text{CH}_3 - \text{Cl}$ (B)  (C)  (D) $\text{MeO} - \text{CH}_2 - \text{Cl}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

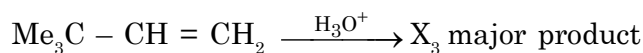
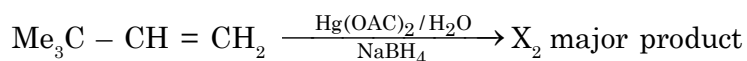
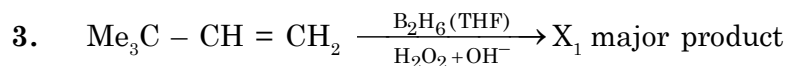
+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- In ZnS (density = 4 gm/ml) unit cell all following manipulations are done -
 (i) All ions along one body diagonal are removed
 (ii) After step (i) one Zn^{2+} is swapped by face centered s^{-2} ion.
 Find density of final unit cell if volume of unit cell remain unchanged [$\text{Zn} = 65.4$, $\text{S} = 32$].
- Find the total number of molecules/ions in which total number of antibonding electrons are more than the total number of bonding electrons?
 C_2 , B_2 , $\text{O}_2^\ominus$, N_2 , F_2 , Li_2 , $\text{N}_2^\oplus$, $\text{N}_2^\ominus$, O_2

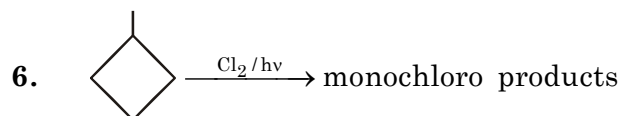
Space for Rough Work



How many options are correct.

- (i) X_1 is 1° alcohol
 - (ii) X_2 is 2° alcohol
 - (iii) X_3 is 3° alcohol
 - (iv) X_3 is formed through carbocation intermediate
 - (v) X_2 is Markovnikov produce
 - (vi) X_1 is anti Markovnikov product.
4. In any polytropic process, molar heat capacity of ideal gas is ' $\text{Cp} - 2\text{R}$ ', calculate work done (in kcal) if one mole of ideal gas is compressed from 300K, 2L to 0.5 litre. [$\text{R} = 2\text{cal/K-mole}$]
5. Consider the following statements.
- (i) Total number of oxygen atoms participate in ring formation in $[\text{Co}(\text{EDTA})]^{-1} = \text{'X'}$
 - (ii) Total number of five membered rings are in $[\text{Co}(\text{dien})(\text{NH}_3)_3]^{+3} = \text{'Y'}$
 - (iii) Maximum number of denticity shown by dmg^{-1} in $[\text{Ni}(\text{DMG})_2] = \text{'Z'}$

then find the value of $\frac{\text{X} + \text{Y}}{\text{Z}} = ?$



Identify total number of monochloro products?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If $\tan^{-1}(x+1) + \cot^{-1}(x-1) = \sin^{-1}\frac{4}{5} + \cos^{-1}\frac{3}{5}$, then possible value of x is
 (A) $4\sqrt{\frac{3}{7}}$ (B) $4\sqrt{\frac{7}{3}}$ (C) $-4\sqrt{\frac{7}{3}}$ (D) $6\sqrt{7}$
2. Given that $f(x)$ is a function defined on $\mathbb{R}$, satisfying $f(1) = 1$ and for any $x \in \mathbb{R}$, $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = f(x) + 5 - x$, then $g(2021)$ is equal to-
 (A) 3 (B) 4 (C) 5 (D) 6
3. If $[x]^3 + [y]^3 + [z]^3 - 3[x][y][z] = 4$ where $x, y, z > 0$ then minimum possible value of $(x+y+z)$ is (where $[.]$ denotes greatest integer function)
 (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

4. Let $(1 + \sqrt{2})^n = A_n + B_n \sqrt{2}$, where A_n and B_n are rational numbers and $n \in \mathbb{N}$, then the value of $\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$ is equal to -

(A) $\left[\lim_{x \rightarrow 0} \frac{\sin x}{x} \right]$

(B) $\lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right]$

(C) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{|\cos x - \sin x|}{\left| x - \frac{\pi}{4} \right|}$

(D) $\left[\lim_{x \rightarrow 0} \frac{\sin \sqrt{2} x}{x} \right]$

(where $[.]$ denotes greatest integer function)

5. If $n \in \mathbb{N}$ and $f(x) = ({}^nC_0 x + 1)({}^nC_1 x + 2)({}^nC_2 x + 3) \dots ({}^nC_n x + n + 1)$ and $f'(0) = \frac{3}{2}(7!)$, then n is equal to

(A) 3

(B) 4

(C) 5

(D) 6

6. Let the function p , q and r defined by $p(x) = x^{\ell n x}$, $q(x) = (\ell n x)^x$, $r(x) = \log_{e^x} x$, then the value of $\frac{p'(e^{-1}) \cdot r'(e^2)}{q'(e)}$ is (where y' denotes the derivative of y)

(A) $\frac{1}{e}$

(B) $\frac{1}{e^2}$

(C) $\frac{2}{e}$

(D) $\frac{2}{e^2}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Identify the **CORRECT** statements-
- (A) Sum of two periodic functions may or maynot be a periodic function
 (B) A differentiable function in $\mathbb{R}$ will always be continuous in $\mathbb{R}$.
 (C) A continuous function whose domain is in a closed interval is always bounded
 (D) In $[a,b]$ for a function $f(x)$, if $f(a) \cdot f(b) < 0$, then there exists atleast one root of $f(x) = 0$ in (a,b) .
8. If $f(x) = \lim_{n \rightarrow \infty} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 3}$, then which of the following statement(s) holds good ?
- (A) $f(x)$ is continuous at $x = 1$ (B) $f(x)$ is discontinuous at $x = 1$
 (C) $\lim_{x \rightarrow 2} f(x)$ exists finitely (D) $\lim_{x \rightarrow 2} f(x)$ does not exist

Space for Rough Work

9. If $f(x) = \begin{cases} \cos^{-1}(\cos x); & |x| < \pi \\ \pi \log_{\pi} |x|; & |x| \geq \pi \end{cases}$, then correct statement(s) is/are -
 (A) $f(x)$ is not differentiable at three points.
 (B) $f'(\pi^+) + f'(-\pi^-) = 0$
 (C) $f'(\pi^+) = f'(-\pi^-)$
 (D) $f'(0^+) + f'(0^-) = 0$
10. The function $f(x) = (\cos x) [\cos x]$ in $(0, 2\pi)$ - (where $[.]$ denotes GIF)
 (A) is discontinuous at two points (B) is non differentiable at two points
 (C) has range $(0, 1]$ (D) is such that $f(f(x)) + 1 = 0$ has no real roots.
11. Given $\lim_{x \rightarrow \alpha} \frac{\sin([\ x^3] - [x]^3)}{(x - \alpha)^3}$ exists finitely, then the value of α can be (where $\alpha \in I$ & $[.]$ denotes greatest integer function)-
 (A) 0 (B) 1 (C) -1 (D) 2
12. Let $f(x) = \begin{cases} \min\{\sin^{-1} x, \cos^{-1} x\} & ; \quad 0 \leq x \leq 1 \\ \max\{\sin^{-1} x, \cos^{-1} x\} & ; \quad -1 \leq x < 0 \end{cases}$ and $f(x) = \frac{1}{2}$, $f(x) = \frac{1}{4}$ & $f(x) = 2$ has number of solution(s) $\lambda_1, \lambda_2, \lambda_3$ respectively, then
 (A) $\lambda_1 = 2$ (B) $\lambda_2 = 1$ (C) $\lambda_2 = 2$ (D) $\lambda_3 = 1$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 4\}$. The total number of functions $f : A \rightarrow B$ such that $f(x) \neq x \forall x \in A$ and range of $f(x)$ contains exactly three elements, is
- The number of values of x , $x \in [-2, 3]$ where $f(x) = [x^2] \sin(\pi x)$ is discontinuous, is (where $[.]$ denotes greatest integer function)-
- Number of points where $f(x) = |(2^x - 1)(3^x - 3)x(x - 1)(x - 2)|$ is not differentiable

Space for Rough Work

4. Let $S_n = \sum_{k=0}^n \tan^{-1} \frac{49}{49k^2 + 63k + 57}$ then $\cot\left(\lim_{n \rightarrow \infty} S_n\right)$ is equal to
5. Let $f(x)$ be a twice differentiable function satisfying $xf(x^2) + (x^2 + 1)f(x) = x$, $\forall x \in \mathbb{R}$ then the value of $|f''(1)|$ is
6. Let $f(x) = \begin{cases} ax & ; x < 2 \\ ax^2 - bx + 3 & ; x \geq 2 \end{cases}$ and ' f ' is differentiable $\forall x \in \mathbb{R}$, then $(a + b)$ is equal to -

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator